

Class 10 · Practice Test (Maths and Science)

Time: 1 hour · Maximum marks: 26 · Answer all questions. Show your working.

Q1.	Two dice are rolled simultaneously. The probability of getting number less than 4 on each die is: (A) $\frac{1}{4}$ (B) $\frac{1}{9}$ (C) $\frac{1}{36}$ (D) $\frac{1}{6}$	[1]
Q2.	Assertion (A): The apparent depth of a coin in water is less than its real depth. Reason (R): Light bends away from the normal when it travels from water to air. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, and R is not the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true.	[1]
Q3.	An electric kettle is rated 750 W; 220 V. Can this kettle be used in a circuit which has a fuse of current rating 3 A? Give reason for your answer.	[2]
Q4.	ABCD is a trapezium in which AB is parallel to DC and its diagonals intersect each other at the point O. Show that $AO/BO = CO/DO$.	[2]
Q5.	(a) Define one volt potential difference between two points in an electric field. [1] (b) Draw a schematic diagram of an electric circuit of a cell of 1.5 V, $5\ \Omega$ and $10\ \Omega$ resistor and a plug key, all connected in series. Calculate the current drawn from the cell when the key is closed. [2]	[3]
Q6.	Ramya performs an experiment in the following two steps: Step 1: Heats ethanol with conc. H_2SO_4 at $170^\circ C$ to get product A. Step 2: Passes hydrogen gas over product A, using Ni as a catalyst and gets product B. (i) Name the products A and B. [1] (ii) Write the reactions involved in both the steps. [2]	[3]
Q7.	Write one function each of the following, in a seed: (i) Seed coat [$\frac{1}{2}$] (ii) Cotyledon [$\frac{1}{2}$] (iii) Radicle [$\frac{1}{2}$] (iv) Plumule [$\frac{1}{2}$]	[2]
Q8.	A solid wooden toy is in the form of a hemisphere surmounted by a cone of the same radius. The radius of hemisphere is 3.5 cm and the total wood used in making the toy is $166\frac{5}{6}\text{ cm}^3$. Find the height of the conical part. Also, find the cost of painting the hemispherical part of the toy at the rate of ₹ 15 per cm^2 . (Take $\pi = \frac{22}{7}$.)	[5]

- Q9. Case study: Acid-base indicators can be used to distinguish between an acid and a base. Universal indicator, a mixture of several indicators, shows different colours at different concentrations of acids and bases, thereby indicating their pH on the pH scale of 0 - 14. The pH of a solution is measured by pH paper, which is a paper impregnated with a universal indicator. [4]
- Answer the following questions:
- (a) Solution P is a strong acid while solution Q is a strong base. On the pH scale, where would you place the solutions P and Q? [1]
 - (b) A solution has a pH of 7. Name a compound you would use to (i) increase its pH, and (ii) decrease its pH. [1]
 - (c) When the pH of a solution is decreased from 4 to 2, what effect does it produce on its hydronium ion concentration? State the colour change shown by the pH paper. [2]
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- Q10. Given that $\sqrt{3}$ is irrational number, prove that $(2 + 3\sqrt{3})$ is an irrational number. [3]
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